
Cross Points and Markers Influence Analysis

Automated Run Report

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Run Configuration

Table 1: Runtime parameters loaded from the XML configuration file.

Parameter	Value
Input path	data/input/all_visits_percentage_variation_cleaned.csv
Output path	data/output
Detail output file	cpm_ia_detail.csv
Aggregated output file	cpm_ia_aggregated.csv
Threshold θ (%)	300.0
Rise tolerance ε (%)	5.0
Max descent window $n_{\max}$	10

Dataset Overview

Table 2: Input dataset dimensions after schema validation and ingestion (Module 02).

Metric	Count
Unique visits	8577
Unique anatomical points	62
Unique markers across all visits	2237
(Visit, point) series analysed	84042
Positive (visit, point) series	15912 (18.9 %)
Descent windows truncated at $n_{\max}$	0 (0.0 %)

Algorithm Description

The CPM-IA component analyses whether excitation events detected in bioelectrical surface current signals systematically influence the signal values recorded at subsequent measurement markers. The pipeline processes long-format tabular measurements through eight sequential analytical modules (M03–M10), each producing typed, immutable outputs consumed by the next stage.

Input Data

The component ingests a single long-format CSV file (UTF-8, comma-separated). Each row encodes one (visit, anatomical point, marker) measurement triplet. The required columns are:

- **Visit identifier** — unique alphanumeric code for the acquisition session.
- **Marker identifier** — unique code identifying the stimulus or measurement marker.
- **Anatomical point identifier** — code identifying the electrode or body-surface acquisition site.
- **Percentage variation** — raw signal change relative to a reference value, expressed as a floating-point percentage (%).

Additional visit-level metadata columns (e.g. acquisition date) are preserved in the detail output without modification.

Configuration Parameters

Three parameters govern the analysis; all are loaded exclusively from the XML configuration file at run time and are never hard-coded:

- **Threshold $\theta = 300.0\%$:** a marker is classified as a *positive excitation* when its percentage variation strictly exceeds θ .
- **Rise tolerance $\varepsilon = 5.0\%$:** the maximum increase permitted between two consecutive values inside the descent window; a rise strictly exceeding ε closes the window.
- **Maximum window length $n_{\max} = 10$:** the descent window is forcibly closed after $n_{\max}$ markers even when no rise exceeding ε has occurred (truncated window).

Processing Steps

M03 — Marker Sequence Reconstruction. For each visit independently, the canonical marker detection order is derived from the first-appearance order of marker identifiers in the input file’s rows for that visit. This per-visit temporal ordering is immutable for the entire run: no sorting, grouping, or reordering of markers is performed after this step. Each (visit, anatomical point) pair is represented as an ordered list of percentage variation values aligned to its visit’s detection order; positions where a marker belongs to the visit protocol but was not measured at a specific point receive a NaN sentinel value.

M04 — First-Positive Trigger Detection. Each (visit, point) sequence is scanned in temporal detection order. The first marker whose percentage variation *strictly exceeds* θ is designated the *excitation anchor* — the onset of the excitation event. The anchor is immutable for the run: once assigned, it cannot be reassigned even if a later marker has a higher value. Series in which no marker exceeds θ are flagged as “no cross-marker effect” and are excluded from all subsequent analytical modules (M06–M09).

M05 — Positive Census. An anatomical point is classified as *positive* if at least one of its visits produced an anchor. The census counts positive and total unique anatomical points across the dataset.

M06 — Descent Window Determination. Starting from the anchor (step $i = 0$), the window extends to successive markers as long as the signal does not rise by more than ε between consecutive steps. Formally, the window closes at the first step $i \geq 1$ where $v_i - v_{i-1} > \varepsilon$. The window is also closed when it reaches $n_{\max}$ markers (truncated window). The anchor is always included as the first element of the window.

M07 — Descent Slope (OLS Linear Regression). An ordinary least-squares (OLS) linear regression is fitted to the L descent window values $y_0, y_1, \dots, y_{L-1}$ at integer step positions $x_i = i$, where $x_0 = 0$ corresponds to the anchor. The closed-form slope estimator is:

$$\hat{\beta}_1 = \frac{L \sum_{i=0}^{L-1} i y_i - \sum_{i=0}^{L-1} i \cdot \sum_{i=0}^{L-1} y_i}{L \sum_{i=0}^{L-1} i^2 - \left(\sum_{i=0}^{L-1} i \right)^2}$$

The slope $\hat{\beta}_1$ (in $\% \text{step}^{-1}$) quantifies the average linear rate of change of the signal across the descent window. A window of length $L = 1$ yields an undefined slope, reported as missing.

M08 — Descent Descriptors. Three scalar shape descriptors are computed for each positive (visit, point) series:

- **Descent depth** = $v_{\text{anchor}} - \min(v_0, \dots, v_{L-1})$, in %: total signal drop from the anchor to the lowest point in the window.
- **Descent length** = L : number of markers in the window, including the anchor.
- **Mean per-step decrease** = $\text{depth} / (L - 1)$, in % step^{-1} : average drop per marker step, normalising depth by window duration. Undefined for $L = 1$.

M09 — Variance Indicator. Sample variance (denominator $n - 1$, i.e. `ddof` = 1) is computed separately on two signal segments:

- **Before-segment:** all sequence values at positions before the anchor index, after removing NaN entries.
- **After-segment:** the descent window values $v_0, \dots, v_{L-1}$, after removing NaN entries.

A segment with fewer than two clean values yields an undefined variance (reported as missing). The **variance ratio** = $\sigma_{\text{after}}^2 / \sigma_{\text{before}}^2$ is computed when both variances are defined and $\sigma_{\text{before}}^2 > 0$. A ratio > 1 indicates that signal variability *increases* after the excitation onset; a ratio < 1 indicates it *decreases*.

M10 — Output Consolidation. Results from M04–M09 are assembled into two tidy output files:

- **Detail file** (`cpm_ia_detail.csv`): one row per (visit, anatomical point) pair, containing all analytical outputs alongside visit metadata.
- **Aggregated file** (`cpm_ia_aggregated.csv`): one row per anatomical point, containing visit counts, positive prevalence, and median descent descriptors computed over positive visits only.

Output Artefacts

The pipeline produces three artefacts in the output directory `data/output`:

1. The per-(visit, point) detail CSV (M10).
2. The per-point aggregation CSV (M10).
3. This automated PDF run report (M11), generated from a filled $\text{\LaTeX}$ template and compiled with `pdflatex` in two passes to resolve table-of-contents page numbers.

No output file is ever silently overwritten: the pipeline raises an error if a target file already exists (guard G-10).

Per-Point Results

Of the **84042** (visit, point) series analysed, **15912 (18.9%)** yielded at least one excitation event above the threshold $\theta = 300.0\%$. **62** of **62** unique anatomical points have at least one positive visit. All tables in this section are sorted alphabetically by point identifier.

Visit Counts and Positive Prevalence

A visit is *positive* for a given anatomical point when at least one marker in that (visit, point) series has a percentage variation strictly exceeding $\theta = 300.0\%$ in temporal detection order (Module M04). All other visits are classified as “no cross-marker effect” for that point.

The **positive prevalence** is the proportion of total visits that are positive:

$$\text{Prevalence (\%)} = \frac{\text{Positive Visits}}{\text{Total Visits}} \times 100$$

A high prevalence indicates that the excitation event is consistently reproduced at a given point across visits; a low prevalence indicates that the event is rare or confined to specific sessions.

Anatomical Point	Total Visits	Positive Visits	Prevalence (%)
LF - 1 - c [43]	902	73	8.1
LF - 1 - e [32]	1801	365	20.3
LF - 1 - i [31]	1205	210	17.4
LF - 2 - c [44]	1137	178	15.7
LF - 2 - e [34]	1547	254	16.4
LF - 2 - i [33]	994	103	10.4
LF - 3 - c [36]	1353	216	16.0
LF - 3 - e [37]	1414	209	14.8
LF - 3 - i [35]	1585	313	19.7
LF - 4 - c [39]	1255	169	13.5
LF - 4 - e [40]	1206	131	10.9
LF - 4 - i [38]	1324	192	14.5
LF - 5 - c [45]	1359	115	8.5
LF - 5 - e [42]	1305	144	11.0
LF - 5 - i [41]	1205	193	16.0
LH - 1 - c [2]	1602	386	24.1
LH - 1 - e [3]	1522	376	24.7
LH - 1 - i [1]	1055	159	15.1
LH - 2 - c [5]	1907	588	30.8
LH - 2 - e [6]	1365	283	20.7
LH - 2 - i [4]	1942	534	27.5
LH - 3 - c [8]	1249	263	21.1
LH - 3 - e [9]	1864	528	28.3
LH - 3 - i [7]	1195	241	20.2
LH - 4 - c [11]	1464	375	25.6
LH - 4 - e [12]	1261	242	19.2
LH - 4 - i [10]	1560	446	28.6
LH - 5 - c [15]	1477	375	25.4
LH - 5 - e [14]	2155	650	30.2
LH - 5 - i [13]	1480	349	23.6
RF - 1 - c [58]	913	82	9.0
RF - 1 - e [47]	1555	268	17.2
RF - 1 - i [46]	1442	179	12.4
RF - 2 - c [59]	990	67	6.8
RF - 2 - e [49]	1421	205	14.4
RF - 2 - i [48]	969	76	7.8
RF - 3 - c [51]	1295	201	15.5
RF - 3 - e [52]	1332	183	13.7
RF - 3 - i [50]	1545	257	16.6
RF - 4 - c [54]	1267	136	10.7

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Anatomical Point	Total Visits	Positive Visits	Prevalence (%)
RF - 4 - e [55]	1167	133	11.4
RF - 4 - i [53]	1421	197	13.9
RF - 5 - c [60]	1248	125	10.0
RF - 5 - e [57]	1362	179	13.1
RF - 5 - i [56]	1140	119	10.4
RH - 1 - c [17]	1625	375	23.1
RH - 1 - e [18]	1371	317	23.1
RH - 1 - i [16]	1124	154	13.7
RH - 2 - c [20]	1854	514	27.7
RH - 2 - e [21]	1164	190	16.3
RH - 2 - i [19]	1870	457	24.4
RH - 3 - c [23]	1135	187	16.5
RH - 3 - e [24]	1463	294	20.1
RH - 3 - i [22]	1182	193	16.3
RH - 4 - c [26]	1397	360	25.8
RH - 4 - e [27]	1207	218	18.1
RH - 4 - i [25]	1518	391	25.8
RH - 5 - c [30]	1359	317	23.3
RH - 5 - e [29]	1924	519	27.0
RH - 5 - i [28]	1645	327	19.9
Thyroid - LH [61]	237	22	9.3
Thyroid - RH [62]	210	10	4.8

Most Frequent Trigger Markers by Anatomical Point

For each anatomical point, the table shows which markers most frequently acted as the *excitation anchor* — the first marker in temporal detection order (Module M03) whose percentage variation strictly exceeded $\theta = 300.0\%$ (Module M04) — across positive visits.

Only markers that acted as anchor in at least **3.5 %** of the positive visits for that point are listed. Markers with a lower share are scattered across sessions with no single dominant trigger and are omitted for readability. All listed marker values exceeded $\theta = 300.0\% > 100\%$ in the visits where they acted as anchor. Anatomical points whose most frequent anchor marker appears in fewer than 3.5 % of positive visits do not appear in this table (no single marker is dominant at those points).

How to read the columns:

- **Trigger Marker:** identifier of the first marker to exceed θ for that point; the value that classified the visit as positive.
- n : number of positive visits for that point in which this marker was the anchor.
- **Share of Pos. Visits (%):** n expressed as a percentage of the total positive visits for that point. Values near 100 % indicate a dominant, reproducible trigger; lower values indicate concurrent triggers across sessions (e.g. due to protocol or biological variability).

Rows are sorted by anatomical point (alphabetical) and, within each point, by descending n .

Anatomical Point	Trigger Marker	n	Share (%)
LF - 1 - c [43]	Acetilcolina [1166]	8	11.0

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Anatomical Point	Trigger Marker	<i>n</i>	Share (%)
LF - 1 - c [43]	Calcium Fluoratum [1151]	8	11.0
LF - 1 - c [43]	Bombesina [1076]	5	6.8
LF - 1 - c [43]	C-RAS [1163]	5	6.8
LF - 1 - c [43]	3-Nitro Tirosina [361]	3	4.1
LF - 1 - c [43]	Insulina [2282]	3	4.1
LF - 1 - c [43]	Profito CIRC [639]	3	4.1
LF - 1 - c [43]	S-100 [1123]	3	4.1
LF - 1 - c [43]	TXA - 2 [551]	3	4.1
LF - 1 - e [32]	Candida albicans [556]	27	7.4
LF - 1 - e [32]	COX-2 [679]	14	3.8
LF - 1 - e [32]	IFN-alfa [2258]	13	3.6
LF - 1 - i [31]	ACTH [893]	16	7.6
LF - 1 - i [31]	8-OHdG [526]	11	5.2
LF - 1 - i [31]	Aspergillus fumigatus [490]	9	4.3
LF - 1 - i [31]	IFN-beta [1452]	8	3.8
LF - 2 - c [44]	Aspen [1136]	15	8.4
LF - 2 - c [44]	Calcium Fluoratum [1151]	13	7.3
LF - 2 - c [44]	Bombesina [1076]	9	5.1
LF - 2 - c [44]	Calcium Phosphoricum [990]	7	3.9
LF - 2 - e [34]	Candida albicans [556]	16	6.3
LF - 2 - e [34]	Aspergillus fumigatus [490]	10	3.9
LF - 2 - e [34]	C-RAS [1163]	9	3.5
LF - 2 - i [33]	Glicina [2]	6	5.8
LF - 2 - i [33]	Calcium Phosphoricum [990]	5	4.9
LF - 2 - i [33]	25toxoplââ sma p [2487]	4	3.9
LF - 2 - i [33]	AGO 1 [686]	4	3.9
LF - 2 - i [33]	Bombesina [1076]	4	3.9
LF - 2 - i [33]	H2O2 [1357]	4	3.9
LF - 3 - c [36]	Calcium Fluoratum [1151]	17	7.9
LF - 3 - c [36]	Bombesina [1076]	14	6.5
LF - 3 - c [36]	AGO 2 [251]	10	4.6
LF - 3 - c [36]	Aspen [1136]	9	4.2
LF - 3 - c [36]	Glutammato [2251]	9	4.2
LF - 3 - e [37]	Calcium Fluoratum [1151]	17	8.1
LF - 3 - e [37]	AGO 1 [686]	13	6.2
LF - 3 - e [37]	Acetilcolina [1166]	10	4.8
LF - 3 - e [37]	AGO 2 [251]	9	4.3
LF - 3 - i [35]	Calcium Fluoratum [1151]	22	7.0
LF - 3 - i [35]	AGO 2 [251]	16	5.1
LF - 3 - i [35]	Acetilcolina [1166]	13	4.2
LF - 3 - i [35]	GABA [1127]	12	3.8
LF - 3 - i [35]	Bombesina [1076]	11	3.5
LF - 4 - c [39]	Aspen [1136]	15	8.9
LF - 4 - c [39]	Acetilcolina [1166]	11	6.5
LF - 4 - c [39]	Calcium Fluoratum [1151]	9	5.3
LF - 4 - e [40]	Aspergillus niger [1267]	6	4.6
LF - 4 - e [40]	Candida albicans [556]	6	4.6
LF - 4 - e [40]	IFN-alfa [2258]	6	4.6
LF - 4 - i [38]	Acetilcolina [1166]	10	5.2
LF - 4 - i [38]	AGO 5 [1073]	9	4.7
LF - 4 - i [38]	Aspen [1136]	9	4.7
LF - 4 - i [38]	AGO 2 [251]	8	4.2
LF - 5 - c [45]	Calcium Fluoratum [1151]	10	8.7
LF - 5 - c [45]	Acetilcolina [1166]	8	7.0
LF - 5 - c [45]	AGO 1 [686]	7	6.1
LF - 5 - c [45]	Aspen [1136]	5	4.3
LF - 5 - c [45]	Leptina [1317]	5	4.3
LF - 5 - e [42]	COX-2 [679]	7	4.9

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Anatomical Point	Trigger Marker	<i>n</i>	Share (%)
LF - 5 - e [42]	Candida albicans [556]	7	4.9
LF - 5 - e [42]	8-OHdG [526]	6	4.2
LF - 5 - e [42]	C-RAS [1163]	6	4.2
LF - 5 - i [41]	Candida albicans [556]	10	5.2
LF - 5 - i [41]	8-OHdG [526]	8	4.1
LF - 5 - i [41]	AGO 5 [1073]	8	4.1
LF - 5 - i [41]	Aspergillus niger [1267]	8	4.1
LF - 5 - i [41]	Aspergillus fumigatus [490]	7	3.6
LH - 1 - c [2]	ACTH [893]	35	9.1
LH - 1 - c [2]	Acetilcolina [1166]	22	5.7
LH - 1 - c [2]	Calcium Fluoratum [1151]	19	4.9
LH - 1 - c [2]	AGO 5 [1073]	15	3.9
LH - 1 - c [2]	AGO 2 [251]	14	3.6
LH - 1 - e [3]	Calcium Fluoratum [1151]	33	8.8
LH - 1 - e [3]	Acetilcolina [1166]	22	5.9
LH - 1 - e [3]	AGO 2 [251]	15	4.0
LH - 1 - e [3]	3-Nitro Tirosina [361]	14	3.7
LH - 1 - i [1]	8-OHdG [526]	8	5.0
LH - 1 - i [1]	AGO 5 [1073]	8	5.0
LH - 1 - i [1]	COX-2 [679]	8	5.0
LH - 1 - i [1]	IFN-gamma [2264]	7	4.4
LH - 1 - i [1]	Acido lattico [1300]	6	3.8
LH - 2 - c [5]	ACTH [893]	48	8.2
LH - 2 - c [5]	Calcium Fluoratum [1151]	31	5.3
LH - 2 - c [5]	AGO 5 [1073]	30	5.1
LH - 2 - c [5]	Acetilcolina [1166]	25	4.3
LH - 2 - e [6]	Acetilcolina [1166]	18	6.4
LH - 2 - e [6]	Calcium Fluoratum [1151]	14	4.9
LH - 2 - e [6]	AGO 1 [686]	13	4.6
LH - 2 - e [6]	Cortisolo [1056]	11	3.9
LH - 2 - e [6]	Rock Rose [435]	11	3.9
LH - 2 - e [6]	ACTH [893]	10	3.5
LH - 2 - e [6]	Glutammato [2251]	10	3.5
LH - 2 - i [4]	Candida albicans [556]	36	6.7
LH - 2 - i [4]	Aspergillus niger [1267]	27	5.1
LH - 2 - i [4]	COX-2 [679]	20	3.7
LH - 3 - c [8]	Acetilcolina [1166]	31	11.8
LH - 3 - c [8]	Calcium Fluoratum [1151]	22	8.4
LH - 3 - c [8]	AGO 2 [251]	13	4.9
LH - 3 - c [8]	Aspen [1136]	11	4.2
LH - 3 - c [8]	Bombesina [1076]	11	4.2
LH - 3 - e [9]	Acetilcolina [1166]	51	9.7
LH - 3 - e [9]	Calcium Fluoratum [1151]	31	5.9
LH - 3 - i [7]	Acetilcolina [1166]	26	10.8
LH - 3 - i [7]	Calcium Fluoratum [1151]	21	8.7
LH - 3 - i [7]	AGO 2 [251]	10	4.1
LH - 4 - c [11]	Acetilcolina [1166]	43	11.5
LH - 4 - c [11]	AGO 2 [251]	24	6.4
LH - 4 - c [11]	Calcium Fluoratum [1151]	22	5.9
LH - 4 - c [11]	Aspen [1136]	18	4.8
LH - 4 - c [11]	Bombesina [1076]	18	4.8
LH - 4 - e [12]	Acetilcolina [1166]	24	9.9
LH - 4 - i [10]	Calcium Fluoratum [1151]	39	8.7
LH - 4 - i [10]	Acetilcolina [1166]	28	6.3
LH - 4 - i [10]	Insulina [2282]	20	4.5
LH - 4 - i [10]	3-Nitro Tirosina [361]	19	4.3
LH - 4 - i [10]	AGO 2 [251]	18	4.0
LH - 4 - i [10]	Pregnenolone [1490]	16	3.6

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Anatomical Point	Trigger Marker	<i>n</i>	Share (%)
LH - 5 - c [15]	Calcium Fluoratum [1151]	27	7.2
LH - 5 - c [15]	AGO 2 [251]	26	6.9
LH - 5 - c [15]	Bombesina [1076]	26	6.9
LH - 5 - c [15]	Acetilcolina [1166]	23	6.1
LH - 5 - c [15]	Aspen [1136]	19	5.1
LH - 5 - e [14]	Candida albicans [556]	33	5.1
LH - 5 - e [14]	Aspergillus niger [1267]	31	4.8
LH - 5 - e [14]	Aspergillus fumigatus [490]	25	3.8
LH - 5 - i [13]	AGO 5 [1073]	29	8.3
LH - 5 - i [13]	3-Nitro Tirosina [361]	14	4.0
RF - 1 - c [58]	Acetilcolina [1166]	11	13.4
RF - 1 - c [58]	Calcium Fluoratum [1151]	8	9.8
RF - 1 - c [58]	Calcium Phosphoricum [990]	6	7.3
RF - 1 - c [58]	Adrenalina [574]	3	3.7
RF - 1 - c [58]	PEtn [2185]	3	3.7
RF - 1 - c [58]	Profito VS - OHI [908]	3	3.7
RF - 1 - c [58]	Psicophyt 2 - Biogroup [210]	3	3.7
RF - 1 - e [47]	Aspergillus fumigatus [490]	18	6.7
RF - 1 - e [47]	AGO 5 [1073]	14	5.2
RF - 1 - e [47]	23tââ`nia sââ²lium v [2488]	11	4.1
RF - 1 - e [47]	Candida albicans [556]	11	4.1
RF - 1 - i [46]	IFN-alfa [2258]	7	3.9
RF - 2 - c [59]	Calcium Fluoratum [1151]	6	9.0
RF - 2 - c [59]	Glutammato [2251]	6	9.0
RF - 2 - c [59]	Acetilcolina [1166]	5	7.5
RF - 2 - c [59]	Dopamina [2259]	4	6.0
RF - 2 - c [59]	Aspen [1136]	3	4.5
RF - 2 - c [59]	Bombesina [1076]	3	4.5
RF - 2 - c [59]	Glutatione-S-Transferasi [723]	3	4.5
RF - 2 - e [49]	Aspergillus fumigatus [490]	10	4.9
RF - 2 - e [49]	Candida albicans [556]	10	4.9
RF - 2 - e [49]	3-Nitro Tirosina [361]	8	3.9
RF - 2 - e [49]	Aspergillus niger [1267]	8	3.9
RF - 2 - i [48]	3-Nitro Tirosina [361]	7	9.2
RF - 2 - i [48]	Acetilcolina [1166]	5	6.6
RF - 2 - i [48]	H2O2 [1357]	4	5.3
RF - 2 - i [48]	23tââ`nia sââ²lium v [2488]	3	3.9
RF - 2 - i [48]	AGO 2 [251]	3	3.9
RF - 2 - i [48]	C-RAS [1163]	3	3.9
RF - 2 - i [48]	Calcium Sulfuricum [701]	3	3.9
RF - 2 - i [48]	IL-2 [2202]	3	3.9
RF - 3 - c [51]	Calcium Fluoratum [1151]	16	8.0
RF - 3 - c [51]	Bombesina [1076]	9	4.5
RF - 3 - c [51]	AGO 5 [1073]	8	4.0
RF - 3 - e [52]	Acetilcolina [1166]	8	4.4
RF - 3 - e [52]	Aspen [1136]	7	3.8
RF - 3 - e [52]	COX-2 [679]	7	3.8
RF - 3 - i [50]	Calcium Fluoratum [1151]	24	9.3
RF - 3 - i [50]	AGO 2 [251]	12	4.7
RF - 3 - i [50]	Acetilcolina [1166]	11	4.3
RF - 3 - i [50]	Aspen [1136]	9	3.5
RF - 3 - i [50]	Bombesina [1076]	9	3.5
RF - 4 - c [54]	Calcium Fluoratum [1151]	12	8.8
RF - 4 - c [54]	Acetilcolina [1166]	6	4.4
RF - 4 - c [54]	Bombesina [1076]	6	4.4
RF - 4 - c [54]	1-HL (1-Hexytol-Lysine) [1368]	5	3.7
RF - 4 - c [54]	Calcium Phosphoricum [990]	5	3.7
RF - 4 - e [55]	Aspergillus fumigatus [490]	7	5.3

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Anatomical Point	Trigger Marker	<i>n</i>	Share (%)
RF - 4 - i [53]	Calcium Fluoratum [1151]	18	9.1
RF - 4 - i [53]	AGO 2 [251]	13	6.6
RF - 4 - i [53]	Dopamina [2259]	8	4.1
RF - 4 - i [53]	Rock Rose [435]	7	3.6
RF - 5 - c [60]	Calcium Fluoratum [1151]	10	8.0
RF - 5 - c [60]	Istamina [2293]	7	5.6
RF - 5 - c [60]	AGO 2 [251]	5	4.0
RF - 5 - c [60]	Acetilcolina [1166]	5	4.0
RF - 5 - c [60]	Leptina [1317]	5	4.0
RF - 5 - c [60]	Pregnenolone [1490]	5	4.0
RF - 5 - e [57]	Candida albicans [556]	11	6.1
RF - 5 - e [57]	Aspergillus fumigatus [490]	9	5.0
RF - 5 - e [57]	ACTH [893]	8	4.5
RF - 5 - e [57]	8-OHdG [526]	7	3.9
RF - 5 - e [57]	COX-2 [679]	7	3.9
RF - 5 - i [56]	Candida albicans [556]	11	9.2
RF - 5 - i [56]	Aspergillus fumigatus [490]	8	6.7
RF - 5 - i [56]	3-Nitro Tirosina [361]	6	5.0
RF - 5 - i [56]	Aspergillus niger [1267]	5	4.2
RH - 1 - c [17]	ACTH [893]	37	9.9
RH - 1 - c [17]	Acetilcolina [1166]	18	4.8
RH - 1 - c [17]	Calcium Fluoratum [1151]	17	4.5
RH - 1 - e [18]	Acetilcolina [1166]	26	8.2
RH - 1 - e [18]	Calcium Fluoratum [1151]	26	8.2
RH - 1 - e [18]	3-Nitro Tirosina [361]	14	4.4
RH - 1 - e [18]	AGO 2 [251]	12	3.8
RH - 1 - e [18]	Aspen [1136]	12	3.8
RH - 1 - i [16]	3-Nitro Tirosina [361]	9	5.8
RH - 1 - i [16]	AGO 5 [1073]	9	5.8
RH - 1 - i [16]	COX-2 [679]	8	5.2
RH - 1 - i [16]	PChoCl [2404]	7	4.5
RH - 1 - i [16]	Aspergillus fumigatus [490]	6	3.9
RH - 1 - i [16]	C-MYC [1487]	6	3.9
RH - 2 - c [20]	ACTH [893]	51	9.9
RH - 2 - c [20]	Beta estradiolo [2272]	21	4.1
RH - 2 - c [20]	AGO 5 [1073]	19	3.7
RH - 2 - c [20]	Acetilcolina [1166]	18	3.5
RH - 2 - e [21]	Acetilcolina [1166]	15	7.9
RH - 2 - e [21]	Calcium Fluoratum [1151]	10	5.3
RH - 2 - e [21]	Adrenalina [574]	9	4.7
RH - 2 - e [21]	Aspen [1136]	8	4.2
RH - 2 - e [21]	AGO 2 [251]	7	3.7
RH - 2 - e [21]	Bombesina [1076]	7	3.7
RH - 2 - e [21]	Dopamina [2259]	7	3.7
RH - 2 - i [19]	Candida albicans [556]	32	7.0
RH - 2 - i [19]	Aspergillus fumigatus [490]	28	6.1
RH - 2 - i [19]	Aspergillus niger [1267]	16	3.5
RH - 2 - i [19]	IFN-gamma [2264]	16	3.5
RH - 3 - c [23]	Acetilcolina [1166]	19	10.2
RH - 3 - c [23]	AGO 2 [251]	10	5.3
RH - 3 - c [23]	Bombesina [1076]	8	4.3
RH - 3 - c [23]	3-Nitro Tirosina [361]	7	3.7
RH - 3 - c [23]	C-RAS [1163]	7	3.7
RH - 3 - e [24]	Acetilcolina [1166]	26	8.8
RH - 3 - e [24]	Calcium Fluoratum [1151]	18	6.1
RH - 3 - e [24]	AGO 2 [251]	16	5.4
RH - 3 - e [24]	3-Nitro Tirosina [361]	11	3.7
RH - 3 - i [22]	Acetilcolina [1166]	20	10.4

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Anatomical Point	Trigger Marker	<i>n</i>	Share (%)
RH - 3 - i [22]	Calcium Fluoratum [1151]	17	8.8
RH - 3 - i [22]	AGO 2 [251]	16	8.3
RH - 3 - i [22]	Bombesina [1076]	10	5.2
RH - 3 - i [22]	AGO 1 [686]	8	4.1
RH - 3 - i [22]	GABA [1127]	7	3.6
RH - 4 - c [26]	Calcium Fluoratum [1151]	29	8.1
RH - 4 - c [26]	AGO 1 [686]	20	5.6
RH - 4 - c [26]	Bombesina [1076]	19	5.3
RH - 4 - c [26]	Acetilcolina [1166]	17	4.7
RH - 4 - c [26]	Aspen [1136]	15	4.2
RH - 4 - e [27]	Acetilcolina [1166]	29	13.3
RH - 4 - e [27]	Calcium Fluoratum [1151]	21	9.6
RH - 4 - e [27]	63 [2097]	10	4.6
RH - 4 - e [27]	Adrenalina [574]	8	3.7
RH - 4 - e [27]	Aspen [1136]	8	3.7
RH - 4 - e [27]	Bombesina [1076]	8	3.7
RH - 4 - i [25]	Acetilcolina [1166]	36	9.2
RH - 4 - i [25]	Calcium Fluoratum [1151]	25	6.4
RH - 4 - i [25]	Aspen [1136]	16	4.1
RH - 5 - c [30]	Acetilcolina [1166]	22	6.9
RH - 5 - c [30]	Calcium Fluoratum [1151]	17	5.4
RH - 5 - c [30]	AGO 2 [251]	16	5.0
RH - 5 - c [30]	Bombesina [1076]	14	4.4
RH - 5 - c [30]	ACTH [893]	12	3.8
RH - 5 - e [29]	Candida albicans [556]	39	7.5
RH - 5 - e [29]	Aspergillus fumigatus [490]	22	4.2
RH - 5 - i [28]	AGO 5 [1073]	16	4.9
RH - 5 - i [28]	Candida albicans [556]	16	4.9
RH - 5 - i [28]	Angiotensina II [561]	13	4.0
Thyroid - LH [61]	Acetilcolina [1166]	2	9.1
Thyroid - LH [61]	Adrenalina [574]	2	9.1
Thyroid - LH [61]	Aspen [1136]	2	9.1
Thyroid - LH [61]	Beta estradiolo [2272]	2	9.1
Thyroid - LH [61]	COX-2 [679]	2	9.1
Thyroid - LH [61]	PChoCl [2404]	2	9.1
Thyroid - LH [61]	5-Hydroxy tryptophol [1197]	1	4.5
Thyroid - LH [61]	Aldosterone [2220]	1	4.5
Thyroid - LH [61]	IFN-alfa [2258]	1	4.5
Thyroid - LH [61]	IGF-1 [1415]	1	4.5
Thyroid - LH [61]	IL-2 [2202]	1	4.5
Thyroid - LH [61]	Marker 10 [2155]	1	4.5
Thyroid - LH [61]	SingoloUno [1650]	1	4.5
Thyroid - LH [61]	Sommatoria [1954]	1	4.5
Thyroid - LH [61]	n08 [1686]	1	4.5
Thyroid - LH [61]	spikeslv [2890]	1	4.5
Thyroid - RH [62]	PChoCl [2404]	3	30.0
Thyroid - RH [62]	ACo [2334]	1	10.0
Thyroid - RH [62]	ACoAPS [2096]	1	10.0
Thyroid - RH [62]	BEs [2313]	1	10.0
Thyroid - RH [62]	Base	1	10.0
Thyroid - RH [62]	Cadmio [751]	1	10.0
Thyroid - RH [62]	PEtn [2185]	1	10.0
Thyroid - RH [62]	SingoloUno [1650]	1	10.0

Descent Shape Medians (Positive Visits Only)

All values in this table are medians computed *exclusively over positive visits* — visits in which an excitation anchor was detected for that point. Points with no positive visits receive a dash

(—) for all descriptor columns. Medians are computed on the raw per-visit values; no smoothing or outlier removal is applied.

How to read and interpret the columns:

- **Med. Slope** (% step⁻¹): median OLS linear regression slope across the descent window (Module M07). A negative value indicates a consistent downward trend in the post-excitation signal; a slope near zero indicates a flat or irregular window. The slope is the rate-of-change estimator that minimises the sum of squared vertical residuals between the fitted line and the window values. Undefined for single-marker windows (—).
- **Med. Depth** (%): median total signal drop from the anchor value to the minimum value in the descent window, $v_{\text{anchor}} - \min_{\text{window}}$ (Module M08). Larger depth indicates a more pronounced excitation response; a depth near zero means the signal barely decreases after the anchor.
- **Med. Length** (marker steps): median number of markers in the descent window, including the anchor (Module M08). A longer window indicates a sustained post-excitation decline before a natural rise exceeding $\varepsilon = 5.0\%$ is observed or before the $n_{\text{max}} = 10$ cap is reached. A length of 1 means the signal rose immediately at the next marker after the anchor.
- **Med. Step Decr.** (% step⁻¹): median mean per-step decrease, computed as depth / (length - 1) (Module M08). This metric normalises the total drop by the window duration, providing a rate-of-descent estimate that is comparable across windows of different lengths. Undefined for single-marker windows (—).

Interpretation guidance. Anatomical points with high positive prevalence (Section 4.1), large median depth, and a well-defined negative slope show a strong and consistent post-excitation descent across visits. Points with low prevalence but large depth when positive are intermittently reactive. A high median length combined with a moderate slope may indicate a gradual, sustained descent rather than a sharp drop. The variance ratio (available in the detail CSV, column `variance_ratio`) complements these descriptors by quantifying whether signal dispersion increases or decreases after the excitation onset.

Anatomical Point	Med. Slope	Med. Depth	Med. Length	Med. Step Decr.
LF - 1 - c [43]	-207.50	185.71	2.0	98.77
LF - 1 - e [32]	-171.43	237.50	2.0	126.92
LF - 1 - i [31]	-161.20	208.57	2.0	114.29
LF - 2 - c [44]	-147.39	187.12	2.0	114.44
LF - 2 - e [34]	-151.92	237.39	2.0	134.39
LF - 2 - i [33]	-160.00	236.36	2.0	132.50
LF - 3 - c [36]	-173.50	231.25	2.0	143.02
LF - 3 - e [37]	-142.86	166.67	2.0	109.09
LF - 3 - i [35]	-173.16	257.14	2.0	141.67
LF - 4 - c [39]	-141.94	192.98	2.0	125.00
LF - 4 - e [40]	-148.08	230.00	2.0	116.00
LF - 4 - i [38]	-171.01	203.57	2.0	125.00
LF - 5 - c [45]	-166.67	178.57	2.0	118.89
LF - 5 - e [42]	-162.50	213.35	2.0	135.16
LF - 5 - i [41]	-146.67	202.50	2.0	115.38
LH - 1 - c [2]	-183.01	228.34	2.0	137.09

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Anatomical Point	Med. Slope	Med. Depth	Med. Length	Med. Step Decr.
LH - 1 - e [3]	-158.62	194.12	2.0	120.29
LH - 1 - i [1]	-160.00	205.88	2.0	112.50
LH - 2 - c [5]	-169.23	211.11	2.0	133.33
LH - 2 - e [6]	-175.00	200.00	2.0	125.00
LH - 2 - i [4]	-149.17	195.78	2.0	115.38
LH - 3 - c [8]	-178.57	180.00	2.0	113.33
LH - 3 - e [9]	-166.09	192.58	2.0	116.90
LH - 3 - i [7]	-152.63	164.71	2.0	105.56
LH - 4 - c [11]	-139.57	173.33	2.0	104.35
LH - 4 - e [12]	-160.66	188.46	2.0	125.32
LH - 4 - i [10]	-162.50	177.12	2.0	114.86
LH - 5 - c [15]	-153.06	233.33	2.0	133.33
LH - 5 - e [14]	-150.00	212.92	2.0	115.80
LH - 5 - i [13]	-166.00	217.86	2.0	127.78
RF - 1 - c [58]	-189.54	188.77	2.0	90.22
RF - 1 - e [47]	-173.37	215.38	2.0	143.21
RF - 1 - i [46]	-150.57	234.78	2.0	108.33
RF - 2 - c [59]	-137.50	162.07	2.0	116.24
RF - 2 - e [49]	-176.47	260.00	2.0	156.25
RF - 2 - i [48]	-188.24	233.33	2.0	159.77
RF - 3 - c [51]	-141.67	206.67	2.0	113.04
RF - 3 - e [52]	-180.00	218.75	2.0	137.25
RF - 3 - i [50]	-168.42	204.35	2.0	136.36
RF - 4 - c [54]	-163.64	200.00	2.0	133.75
RF - 4 - e [55]	-145.65	226.67	2.0	115.79
RF - 4 - i [53]	-156.25	200.00	2.0	125.73
RF - 5 - c [60]	-150.00	235.71	2.0	117.31
RF - 5 - e [57]	-140.55	200.00	2.0	109.52
RF - 5 - i [56]	-168.75	187.50	2.0	142.86
RH - 1 - c [17]	-176.67	246.67	2.0	134.21
RH - 1 - e [18]	-166.29	188.00	2.0	120.00
RH - 1 - i [16]	-163.64	200.00	2.0	132.74
RH - 2 - c [20]	-171.13	231.03	2.0	153.24
RH - 2 - e [21]	-166.67	198.33	2.0	141.18
RH - 2 - i [19]	-152.38	211.11	2.0	127.78
RH - 3 - c [23]	-160.87	190.48	2.0	135.00
RH - 3 - e [24]	-167.50	216.23	2.0	127.22
RH - 3 - i [22]	-166.67	217.39	2.0	115.00
RH - 4 - c [26]	-183.33	183.33	2.0	117.42
RH - 4 - e [27]	-151.67	155.04	2.0	106.97
RH - 4 - i [25]	-170.00	200.00	2.0	120.83
RH - 5 - c [30]	-161.54	175.00	2.0	104.00
RH - 5 - e [29]	-165.38	226.00	2.0	141.67
RH - 5 - i [28]	-162.50	221.05	2.0	145.61
Thyroid - LH [61]	-199.00	54.23	2.0	54.23
Thyroid - RH [62]	-180.00	10.00	1.5	10.00